MATHEMATICS / INDEPENDENT STUDY

MM5020

Mathematics III - Abstract Algebra

Progressive hints

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

Edition 2 · 25 October 2026

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How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give an argument and criteria for self-assessment. Expanded solutions also explain how to find the approach, a tempting dead end, and the role of the hypotheses.

Each unit has twelve core problems (four E, five C, three A) and an optional A-level transfer challenge numbered 13. The challenge asks you to choose a method in a new setting; budget its hour separately. Problem 12 and the challenge have detailed solutions with independently written decisive ideas. Mock estimates total 170 minutes of problem work plus ten minutes for review.

1 Unit 1 · Groups and cyclic structure

MM5020-U1.01 · E · 15 min
Learning objective: U1.1 Group axioms

Show that the even integers form a group under addition, and explain why they do not form a group under multiplication.

Hint 1 — a starting point. List the identity and inverse for addition.

Hint 2 — the key move. The multiplicative identity would have to be one.

MM5020-U1.02 · E · 15 min
Learning objective: U1.1 Cancellation

Solve $axb = c$ in an arbitrary group and prove uniqueness.

Hint 1 — a starting point. Undo the left factor on the left.

Hint 2 — the key move. Multiply first by a^{-1} on the left and then by b^{-1} on the right.

MM5020-U1.03 · E · 15 min
Learning objective: U1.2 Element orders

Find the order and generated subgroup of $[9]$ in $\mathbb{Z}/15\mathbb{Z}$.

Hint 1 — a starting point. List successive multiples or use the gcd formula.

Hint 2 — the key move. The order is $\frac{15}{\gcd(15,9)}$.

MM5020-U1.04 · E · 15 min
Learning objective: U1.2 Subgroup test

Prove that the intersection of two subgroups of a group is a subgroup.

Hint 1 — a starting point. The intersection contains the identity.

Hint 2 — the key move. Apply the one-step subgroup test in both subgroups.

MM5020-U1.05 · C · 30 min
Learning objective: U1.1 Structural identities

Prove that a group in which $x^2 = e$ for every x is abelian.

Hint 1 — a starting point. Every element is its own inverse.

Hint 2 — the key move. Compare $(ab)^{-1}$ with $b^{-1}a^{-1}$.

MM5020-U1.06 · C · 30 min
Learning objective: U1.2 Cyclic subgroups

List all subgroups of $\mathbb{Z}/18\mathbb{Z}$ and their orders.

Hint 1 — a starting point. Subgroups of a cyclic group correspond to divisors of its order.

Hint 2 — the key move. For each divisor d of eighteen, use $\langle [d] \rangle$.

MM5020-U1.07 · C · 30 min
Learning objective: U1.2 Generator arithmetic
Let a have order n. Prove $\langle a^r \rangle = \langle a^{\gcd(n,r)} \rangle$.
Hint 1 — a starting point. Use a Bézout identity for n, r.
Hint 2 — the key move. If $d = un + vr$, then $a^d = (a^r)^v$.

MM5020-U1.08 · C · 30 min
Learning objective: U1.2 Subgroup structure
Prove that the union of two subgroups H, K is a subgroup exactly when one contains the other.
Hint 1 — a starting point. Assume neither contains the other and choose one element from each difference.
Hint 2 — the key move. For $h \in H \setminus K$ and $k \in K \setminus H$, test hk.

MM5020-U1.09 · C · 30 min
Learning objective: U1.1 Noncommutative examples
Let r rotate a square by a quarter turn and s reflect it in a vertical axis. Show $srs = r^{-1}$ and use this to prove the symmetry group is nonabelian.
Hint 1 — a starting point. Follow a vertex through the three motions.
Hint 2 — the key move. Reflection reverses orientation, hence reverses the direction of rotation.

MM5020-U1.10 · A · 50 min
Learning objective: U1.2 Automorphisms
Classify all automorphisms of the additive group $\mathbb{Z}/n\mathbb{Z}$.
Hint 1 — a starting point. A homomorphism is determined by the image of $[1]$.
Hint 2 — the key move. The image must generate the group.

MM5020-U1.11 · A · 50 min
Learning objective: U1.2 Products of cyclic groups
Prove that $C_m \times C_n$ is cyclic exactly when $\gcd(m, n) = 1$.
Hint 1 — a starting point. Find the order of a pair of elements.
Hint 2 — the key move. A pair of generators has order $\operatorname{lcm}(m, n)$.

MM5020-U1.12 · A · 50 min
Learning objective: U1.1 Axioms and finiteness
Let M be a finite nonempty set with an associative operation and a two-sided identity. Assume left and right cancellation. Prove that M is a group.
Hint 1 — a starting point. Multiplication by a fixed element is injective.
Hint 2 — the key move. On a finite set it is surjective, so left and right inverses exist.

MM5020-U1.13 · A · 60 min · Optional challenge

Learning objective: U1 Transfer: A disguised operation

On $X = \mathbb{R}\{-1\}$ define $a \star b = a + b + ab$. Determine whether this is a group, find its identity and inverses, and find all elements of finite order.

Hint 1 — a starting point. Inspect the expression after adding one.

Hint 2 — the key move. Relate $1 + (a \star b)$ to $(1 + a)(1 + b)$.

2 Unit 2 · Cosets and permutations

MM5020-U2.01 · E · 15 min
Learning objective: U2.1 Cosets
List the cosets of $H = \{[0], [4], [8]\}$ in $\mathbb{Z}/12\mathbb{Z}$.
Hint 1 — a starting point. Translate H by small residues.
Hint 2 — the key move. There are $\frac{12}{3} = 4$ cosets.

MM5020-U2.02 · E · 15 min
Learning objective: U2.1 Lagrange
What element orders are possible in a group of order twenty-one according to Lagrange's theorem? Does this alone guarantee an element of order twenty-one?
Hint 1 — a starting point. List the positive divisors.
Hint 2 — the key move. A necessary condition need not be sufficient.

MM5020-U2.03 · E · 15 min
Learning objective: U2.2 Permutation composition
Compute $(123)(23)$ and its sign, composing right to left.
Hint 1 — a starting point. Track the images of one, two, and three.
Hint 2 — the key move. One maps to two, two maps to one, and three is fixed.

MM5020-U2.04 · E · 15 min
Learning objective: U2.2 Cycle order
Find the order and sign of $(1234)(567)$ in S_7.
Hint 1 — a starting point. The cycles are disjoint.
Hint 2 — the key move. Take the least common multiple and multiply their signs.

MM5020-U2.05 · C · 30 min
Learning objective: U2.1 Coset criteria
Prove $aH = bH$ if and only if $a^{-1}b \in H$.
Hint 1 — a starting point. Start with membership of b in aH.
Hint 2 — the key move. Write $b = ah$.

MM5020-U2.06 · C · 30 min
Learning objective: U2.1 Prime order
Prove that a group of prime order has no proper nontrivial subgroup and is cyclic.
Hint 1 — a starting point. Apply Lagrange twice.
Hint 2 — the key move. The subgroup generated by a nonidentity element has order greater than one.

MM5020-U2.07 · C · 30 min
Learning objective: U2.2 Powers of cycles
Describe the disjoint cycles of $(123456)^2$ and $(123456)^3$, and compute their orders.
Hint 1 — a starting point. Adding the exponent modulo six describes the action on positions.
Hint 2 — the key move. The cycles are the orbits of adding two or three.

MM5020-U2.08 · C · 30 min
Learning objective: U2.2 Counting permutations
How many three-cycles are in S_5? How many products of two disjoint transpositions?
Hint 1 — a starting point. Choose supports before choosing arrangements.
Hint 2 — the key move. A three-element support has two cycles; a four-element support has three pairings.

MM5020-U2.09 · C · 30 min
Learning objective: U2.2 Alternating groups
Show every three-cycle is even and every even permutation is a product of three-cycles.
Hint 1 — a starting point. Pair the transpositions in an even factorization.
Hint 2 — the key move. A pair sharing a letter is a three-cycle; a disjoint pair can be split using an intermediate transposition.

MM5020-U2.10 · A · 50 min
Learning objective: U2.2 Minimal permutation degree
Find the least n for which S_n has an element of order fifteen.
Hint 1 — a starting point. Cycle lengths must have least common multiple fifteen.
Hint 2 — the key move. Compare one fifteen-cycle with disjoint cycles of lengths three and five.

MM5020-U2.11 · A · 50 min
Learning objective: U2.1 Indices in a tower
For finite G and $K \leq H \leq G$, prove $[G : K] = [G : H][H : K]$ by constructing coset representatives.
Hint 1 — a starting point. Choose representatives for both stages.
Hint 2 — the key move. Products of representatives give representatives for G/K.

MM5020-U2.12 · A · 50 min
Learning objective: U2.2 Conjugation and cycles
Prove $\tau(a_1 \ldots a_r)\tau^{-1} = (\tau(a_1) \ldots \tau(a_r))$. Deduce that conjugate permutations have the same cycle lengths, and conversely.
Hint 1 — a starting point. Evaluate the conjugate on $\tau(a_i)$.
Hint 2 — the key move. For the converse, choose a bijection matching corresponding cycle positions.

MM5020-U2.13 · A · 60 min · Optional challenge

Learning objective: U2 Transfer: Symmetries with fixed positions

How many permutations of six letters have order two and exactly two fixed letters? Prove that any two are conjugate, and determine how many permutations commute with one of them.

Hint 1 — a starting point. An order-two permutation has only one-cycles and two-cycles.

Hint 2 — the key move. Choose its fixed letters, then partition the remaining four letters into pairs.

3 Unit 3 · Homomorphisms and quotients

MM5020-U3.01 · E · 15 min
Learning objective: U3.1 Kernels
Find the kernel and image of $f : \mathbb{Z} \to \mathbb{Z}/8\mathbb{Z}$, $f(n) = [3n]$.
Hint 1 — a starting point. Three is invertible modulo eight.
Hint 2 — the key move. Its inverse is also three.

MM5020-U3.02 · E · 15 min
Learning objective: U3.1 Normality
Show the kernel of the determinant map $\mathrm{GL}_2(\mathbb{R}) \to \mathbb{R}^\times$ is normal.
Hint 1 — a starting point. Use the homomorphism property of determinants.
Hint 2 — the key move. The kernel consists of matrices of determinant one.

MM5020-U3.03 · E · 15 min
Learning objective: U3.2 Quotient size
Let $|G| = 60$ and $N \lhd G$ have order twelve. Determine $|G/N|$ and its group structure.
Hint 1 — a starting point. Count cosets.
Hint 2 — the key move. A group of order five is cyclic.

MM5020-U3.04 · E · 15 min
Learning objective: U3.1 Homomorphisms
Show a homomorphism $f : G \to H$ satisfies $f(g^n) = f(g)^n$ for every integer n.
Hint 1 — a starting point. Handle positive, zero, and negative exponents separately.
Hint 2 — the key move. Use preservation of inverses for negative exponents.

MM5020-U3.05 · C · 30 min
Learning objective: U3.2 First isomorphism theorem
Identify $\mathbb{R}^\times/\{1, -1\}$ using a natural homomorphism.
Hint 1 — a starting point. Take the absolute value.
Hint 2 — the key move. Its image is the multiplicative group of positive real numbers.

MM5020-U3.06 · C · 30 min
Learning objective: U3.1 Normality
Prove the center $Z(G)$ is a normal subgroup.
Hint 1 — a starting point. Check the subgroup test, then conjugation.
Hint 2 — the key move. Central elements are fixed by every conjugation.

MM5020-U3.07 · C · 30 min
Learning objective: U3.2 Quotient maps

For positive integers m, n, when is $[a]_m \mapsto [a]_n$ well defined? When it is, find its kernel.
Hint 1 — a starting point. The representatives zero and m must agree after mapping.
Hint 2 — the key move. The necessary and sufficient condition is $n|m$.

MM5020-U3.08 · C · 30 min
Learning objective: U3.2 Correspondence

List all subgroups of $\mathbb{Z}/12\mathbb{Z}$ by using subgroups of $\mathbb{Z}$ containing $12\mathbb{Z}$.
Hint 1 — a starting point. All subgroups of $\mathbb{Z}$ are $d\mathbb{Z}$.
Hint 2 — the key move. Containment $12\mathbb{Z} \subseteq d\mathbb{Z}$ means $d|12$.

MM5020-U3.09 · C · 30 min
Learning objective: U3.1 Cyclic quotient by center

Prove that if $G/Z(G)$ is cyclic, then G is abelian.
Hint 1 — a starting point. Express every element using a representative of the quotient generator.
Hint 2 — the key move. Write $x = g^r z$ and $y = g^s w$ with z, w central.

MM5020-U3.10 · A · 50 min
Learning objective: U3.2 Universal property

Prove that a homomorphism $f : G \to H$ factors through G/N exactly when $N \subseteq \ker f$, and prove uniqueness of the factor.
Hint 1 — a starting point. A factor must kill the kernel of the projection.
Hint 2 — the key move. For sufficiency define $\overline{f}(gN) = f(g)$ and check representatives.

MM5020-U3.11 · A · 50 min
Learning objective: U3.2 Abelianization

Let $[G, G]$ be generated by all commutators $aba^{-1}b^{-1}$. Show it is normal, its quotient is abelian, and every map from G to an abelian group factors uniquely through this quotient.
Hint 1 — a starting point. Conjugation takes commutators to commutators.
Hint 2 — the key move. An abelian target kills each commutator.

MM5020-U3.12 · A · 50 min
Learning objective: U3.1 Failure of converse to Lagrange

Prove that A_4 has no subgroup of order six. You may use that three-cycles generate A_4.
Hint 1 — a starting point. A subgroup of order six would have index two.
Hint 2 — the key move. The quotient would have order two, but every three-cycle must map to its identity.

MM5020-U3.13 · A · 60 min · Optional challenge

Learning objective: U3 Transfer: A large cyclic kernel

Let $G = (\mathbb{Z}/12\mathbb{Z}) \times (\mathbb{Z}/18\mathbb{Z})$. Define $f : G \to \mathbb{Z}/6\mathbb{Z}$ by $f(a, b) = a + b$. Determine the kernel and prove that it is cyclic; give a generator and the quotient up to isomorphism.

Hint 1 — a starting point. Check first that changing either representative does not change the sum modulo six.

Hint 2 — the key move. Investigate the order of $(1, -1)$ in the product.

4 Unit 4 · Actions and simple groups

MM5020-U4.01 · E · 15 min
Learning objective: U4.1 Actions
Verify that S_n acts on $\{1, \dots, n\}$ by evaluation.
Hint 1 — a starting point. Check identity and composition.
Hint 2 — the key move. The operation in S_n is function composition.

MM5020-U4.02 · E · 15 min
Learning objective: U4.1 Stabilizers
Find the stabilizer of one in the natural S_4 action and its order.
Hint 1 — a starting point. It permutes the other three letters freely.
Hint 2 — the key move. It is isomorphic to S_3.

MM5020-U4.03 · E · 15 min
Learning objective: U4.2 Conjugacy
Compute $(123)(12)(123)^{-1}$.
Hint 1 — a starting point. Conjugation relabels a cycle.
Hint 2 — the key move. Send one to two and two to three.

MM5020-U4.04 · E · 15 min
Learning objective: U4.2 Simplicity
Which nontrivial finite cyclic groups are simple?
Hint 1 — a starting point. Every subgroup of an abelian group is normal.
Hint 2 — the key move. A cyclic group of composite order has a proper nontrivial subgroup.

MM5020-U4.05 · C · 30 min
Learning objective: U4.1 Kernel of an action
Show the kernel of the homomorphism associated with an action is the intersection of all stabilizers.
Hint 1 — a starting point. Both descriptions say an element fixes every point.
Hint 2 — the key move. Write the two membership conditions explicitly.

MM5020-U4.06 · C · 30 min
Learning objective: U4.1 Conjugate stabilizers
Prove $G_{h \cdot x} = hG_x h^{-1}$.
Hint 1 — a starting point. Apply h^{-1} to the equation defining a stabilizer.
Hint 2 — the key move. The condition is equivalent to $h^{-1}gh$ fixing x.

MM5020-U4.07 · C · 30 min
Learning objective: U4.2 Class equation
Use conjugacy classes to show the center of S_3 is trivial.
Hint 1 — a starting point. List its cycle types.
Hint 2 — the key move. The class sizes are one, three, and two.

MM5020-U4.08 · C · 30 min
Learning objective: U4.2 Prime-power groups
Prove a group of order p^2 is abelian.
Hint 1 — a starting point. The center has order divisible by p.
Hint 2 — the key move. If its order were p, the quotient by the center would be cyclic.

MM5020-U4.09 · C · 30 min
Learning objective: U4.1 Orbit counting
Count two-colorings of four labeled square vertices up to rotation, with reflections excluded.
Hint 1 — a starting point. Use the cyclic rotation group of order four.
Hint 2 — the key move. The four rotations fix $16, 2, 4, 2$ colorings respectively.

MM5020-U4.10 · A · 50 min
Learning objective: U4.2 Simple prime-power groups
Classify finite simple groups whose order is a power of a prime.
Hint 1 — a starting point. Their center is a nontrivial normal subgroup.
Hint 2 — the key move. Simplicity forces the center to be the whole group.

MM5020-U4.11 · A · 50 min
Learning objective: U4.1 Coset actions
Let H have finite index n in G. Show there is a normal subgroup $N \subseteq H$ with $[G : N] \leq n!$.
Hint 1 — a starting point. Let G act on its left cosets of H.
Hint 2 — the key move. Use the kernel of the resulting map to S_n.

MM5020-U4.12 · A · 50 min
Learning objective: U4.2 Faithful actions of simple groups
Prove every nontrivial action of a simple group is faithful. Deduce that A_5 cannot act nontrivially on a set of four elements.
Hint 1 — a starting point. The kernel of an action is normal.
Hint 2 — the key move. A faithful action on four points would embed A_5 into S_4.

MM5020-U4.13 · A · 60 min · Optional challenge

Learning objective: U4 Transfer: Four-position ornaments

An ornament has four positions at the vertices of a square, each colored with one of three colors. Two colorings are identified if a rotation or reflection of the square relates them. How many ornaments are there? Explain why simply dividing the number of colorings by eight fails.

Hint 1 — a starting point. Group the eight square symmetries by their cycle structures on vertices.

Hint 2 — the key move. A coloring fixed by a permutation must be constant on each cycle.

5 Unit 5 · Rings and their quotients

MM5020-U5.01 · E · 15 min
Learning objective: U5.1 Units
Find every unit in $\mathbb{Z}/10\mathbb{Z}$ and its inverse.
Hint 1 — a starting point. A residue is a unit exactly when it is coprime to ten.
Hint 2 — the key move. Test the residues one, three, seven, and nine.

MM5020-U5.02 · E · 15 min
Learning objective: U5.1 Zero divisors
Give nonzero matrices $A, B \in M_2(\mathbb{Q})$ with $AB = 0$.
Hint 1 — a starting point. Use projections onto different coordinate axes.
Hint 2 — the key move. Take diagonal matrices with complementary single nonzero entries.

MM5020-U5.03 · E · 15 min
Learning objective: U5.2 Ideals
Show $6\mathbb{Z}$ is an ideal of $\mathbb{Z}$, and describe the identity of its quotient.
Hint 1 — a starting point. Check additive subgroup and absorption.
Hint 2 — the key move. Multiplying a multiple of six by any integer preserves that form.

MM5020-U5.04 · E · 15 min
Learning objective: U5.1 Quaternions
Find the inverse of $q = 1 + 2i - j$.
Hint 1 — a starting point. Use quaternion conjugation.
Hint 2 — the key move. Its squared norm is six.

MM5020-U5.05 · C · 30 min
Learning objective: U5.2 Quotient criteria
Determine whether (5) and (6) are prime or maximal in $\mathbb{Z}$.
Hint 1 — a starting point. Inspect the quotient rings.
Hint 2 — the key move. A prime-order residue ring is a field; modulo six there are zero divisors.

MM5020-U5.06 · C · 30 min
Learning objective: U5.2 Evaluation
Show $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$.
Hint 1 — a starting point. Evaluate a polynomial at i.
Hint 2 — the key move. Divide by $x^2 + 1$ to identify the kernel.

MM5020-U5.07 · C · 30 min
Learning objective: U5.1 Characteristic
Prove a nonzero domain has characteristic zero or a prime.
Hint 1 — a starting point. A composite characteristic factors.
Hint 2 — the key move. Translate the integer factorization into a zero product in the ring.

MM5020-U5.08 · C · 30 min
Learning objective: U5.2 Ideals and units
Prove an ideal contains a unit if and only if it is the whole ring. Deduce that a field has exactly two ideals.
Hint 1 — a starting point. Multiply the unit by its inverse.
Hint 2 — the key move. An ideal containing one contains every element.

MM5020-U5.09 · C · 30 min
Learning objective: U5.2 Noncommutative ideals
Show the matrices with zero second column form a left ideal of $M_2(K)$ but not a two-sided ideal.
Hint 1 — a starting point. Left multiplication preserves a zero column.
Hint 2 — the key move. Multiply E_{11} on the right by E_{12}.

MM5020-U5.10 · A · 50 min
Learning objective: U5.2 Chinese remainder theorem
Prove $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ for coprime positive m, n.
Hint 1 — a starting point. Use simultaneous reduction and a Bézout identity.
Hint 2 — the key move. The kernel is $mn\mathbb{Z}$; use Bézout to construct any prescribed pair of residues.

MM5020-U5.11 · A · 50 min
Learning objective: U5.2 Matrix ring ideals
Prove every two-sided ideal of $M_{n(K)}$, for a field K, is zero or the whole ring.
Hint 1 — a starting point. Start with a nonzero entry of a matrix in the ideal.
Hint 2 — the key move. Multiplication by matrix units isolates that entry in any desired position.

MM5020-U5.12 · A · 50 min
Learning objective: U5.1 Division rings
Prove quaternion conjugation reverses products and deduce multiplicativity of $N(q) = q\overline{q}$.
Hint 1 — a starting point. It is enough to verify the reversal on the basis elements and extend bilinearly.
Hint 2 — the key move. Use $\overline{qr} = \overline{rq}$ and centrality of real norms.

MM5020-U5.13 · A · 60 min · Optional challenge

Learning objective: U5 Transfer: Coordinates hidden in residues

Find all e in $\mathbb{Z}/12\mathbb{Z}$ satisfying $e^2 = e$. Construct an explicit ring isomorphism from $\mathbb{Z}/12\mathbb{Z}$ to $(\mathbb{Z}/3\mathbb{Z}) \times (\mathbb{Z}/4\mathbb{Z})$ and use it to explain your list.

Hint 1 — a starting point. Compute the two residues of a class modulo twelve.

Hint 2 — the key move. The classes of four and nine isolate the two coordinates.

6 Unit 6 · Polynomials and factorization

MM5020-U6.01 · E · 15 min
Learning objective: U6.1 Division
Divide $x^3 + 2x + 1$ by $x^2 + 1$ over $\mathbb{Q}$.
Hint 1 — a starting point. Cancel the leading term first.
Hint 2 — the key move. Subtract $x(x^2 + 1)$.

MM5020-U6.02 · E · 15 min
Learning objective: U6.1 Bézout
Find integers u, v with $30u + 21v = 3$.
Hint 1 — a starting point. Run the Euclidean algorithm.
Hint 2 — the key move. Use $3 = 21 - 2(30 - 21)$.

MM5020-U6.03 · E · 15 min
Learning objective: U6.2 Irreducibility
Prove $x^3 - 5$ is irreducible over $\mathbb{Q}$.
Hint 1 — a starting point. Use a prime dividing the constant term.
Hint 2 — the key move. Eisenstein applies at five.

MM5020-U6.04 · E · 15 min
Learning objective: U6.1 Roots
Factor $x^2 + 1$ over $\mathbb{F}_5$.
Hint 1 — a starting point. Find a square equal to negative one.
Hint 2 — the key move. Two squared equals four modulo five.

MM5020-U6.05 · C · 30 min
Learning objective: U6.1 Modular inverses
Find the inverse of $x + 1$ in $\mathbb{Q}[x]/(x^2 + 1)$.
Hint 1 — a starting point. Multiply by $x - 1$.
Hint 2 — the key move. The product is $x^2 - 1$, which becomes negative two.

MM5020-U6.06 · C · 30 min
Learning objective: U6.2 Reduction test
Prove $x^3 + x + 1$ is irreducible over $\mathbb{Q}$ by reducing modulo two.
Hint 1 — a starting point. A cubic over a field is reducible exactly when it has a root.
Hint 2 — the key move. Evaluate at the only two elements of $\mathbb{F}_2$.

MM5020-U6.07 · C · 30 min
Learning objective: U6.2 Failure outside PIDs
Show 2 is irreducible but not prime in $\mathbb{Z}\left[\sqrt{-5}\right]$.
Hint 1 — a starting point. Use the norm $N\left(a+b\sqrt{-5}\right)=a^2+5b^2$.
Hint 2 — the key move. There is no element of norm two, but $\left(1+\sqrt{-5}\right)\left(1-\sqrt{-5}\right)=6$.

MM5020-U6.08 · C · 30 min
Learning objective: U6.1 Principal ideals
Show $K[x]/(f)$ is a field if and only if f is irreducible, assuming f is nonconstant.
Hint 1 — a starting point. Use maximal ideals or Bézout.
Hint 2 — the key move. For irreducible f, every polynomial not divisible by f is coprime to it.

MM5020-U6.09 · C · 30 min
Learning objective: U6.2 Primitive products
Show the product of two primitive integer polynomials is primitive.
Hint 1 — a starting point. Suppose a prime divides every coefficient of the product.
Hint 2 — the key move. Reduce modulo that prime and use that $\mathbb{F}_{p[x]}$ is a domain.

MM5020-U6.10 · A · 50 min
Learning objective: U6.2 Shifted Eisenstein
For prime p, prove $1+x+\ldots+x^{p-1}$ is irreducible over $\mathbb{Q}$.
Hint 1 — a starting point. Substitute $x=y+1$.
Hint 2 — the key move. The transformed polynomial is $\frac{(y+1)^p-1}{y}$.

MM5020-U6.11 · A · 50 min
Learning objective: U6.1 Polynomial Chinese remainder
If $f,g\in K[x]$ are coprime, prove $K[x]/(fg)\cong K[x]/(f)\times K[x]/(g)$.
Hint 1 — a starting point. Use simultaneous reduction and polynomial Bézout.
Hint 2 — the key move. Construct a preimage of $([a],[b])$ using $uf+vg=1$.

MM5020-U6.12 · A · 50 min
Learning objective: U6.2 Nonprincipal ideals
Prove $\mathbb{Z}[x]$ is a UFD but not a PID.
Hint 1 — a starting point. Apply Gauss's theorem, then study $(2,x)$.
Hint 2 — the key move. A generator of $(2,x)$ would divide both two and x.

MM5020-U6.13 · A · 60 min · Optional challenge

Learning objective: U6 Transfer: Two quartics without rational roots

Determine which of $x^4 + 1$ and $x^4 + 4$ are irreducible over $\mathbb{Q}$. Give complete factorizations into irreducibles in $\mathbb{Z}[x]$ and explain why testing rational roots does not settle either question.

Hint 1 — a starting point. For the first polynomial, a translation of the variable may reveal a useful coefficient pattern.

Hint 2 — the key move. For the second, add and subtract a square; for the first examine $(x + 1)^4 + 1$.

7 Unit 7 · Fields as vector spaces

MM5020-U7.01 · E · 15 min
Learning objective: U7.1 Fractions

Prove $\frac{a}{b} = 0$ in the fraction field of a domain exactly when $a = 0$.
Hint 1 — a starting point. Compare with the fraction $\frac{0}{1}$.
Hint 2 — the key move. Use the cross-multiplication definition.

MM5020-U7.02 · E · 15 min
Learning objective: U7.2 Minimal polynomials

Find the minimal polynomial of $\sqrt{7}$ over $\mathbb{Q}$ and a basis of its generated field.
Hint 1 — a starting point. Use the polynomial $x^2 - 7$.
Hint 2 — the key move. It is irreducible by Eisenstein at seven.

MM5020-U7.03 · E · 15 min
Learning objective: U7.1 Tower law

If $[L : K] = 3$ and $[M : L] = 4$, find $[M : K]$. Can an intermediate field between K and M have degree five over K?
Hint 1 — a starting point. Multiply the degrees.
Hint 2 — the key move. An intermediate degree must divide the total degree.

MM5020-U7.04 · E · 15 min
Learning objective: U7.2 Power-basis arithmetic

With $\alpha^3 = 2$, simplify $\alpha^5 + \alpha^4$ in the power basis.
Hint 1 — a starting point. Reduce each exponent using $\alpha^3 = 2$.
Hint 2 — the key move. The remaining powers should be below three.

MM5020-U7.05 · C · 30 min
Learning objective: U7.2 Inverses

Find $\left(2 + \sqrt{3}\right)^{-1}$ in $\mathbb{Q}\left(\sqrt{3}\right)$ and explain why every nonzero $a + b\sqrt{3}$ has an inverse in this field.
Hint 1 — a starting point. Multiply by the conjugate expression.
Hint 2 — the key move. The denominator is $a^2 - 3b^2$.

MM5020-U7.06 · C · 30 min
Learning objective: U7.2 Algebraicity

Prove every element of a finite extension is algebraic.
Hint 1 — a starting point. Use too many powers in a finite-dimensional vector space.
Hint 2 — the key move. If the degree is d, the $d + 1$ elements $1, \alpha, \ldots, \alpha^d$ are dependent.

MM5020-U7.07 · C · 30 min
Learning objective: U7.2 Change of generator
Find the minimal polynomial of $1+\sqrt{2}$ over $\mathbb{Q}$.
Hint 1 — a starting point. Isolate the square root and square.
Hint 2 — the key move. A quadratic with no rational root is irreducible.

MM5020-U7.08 · C · 30 min
Learning objective: U7.1 Fraction universal property
Explain why the reduction map $\mathbb{Z} \to \mathbb{F}_5$ cannot extend to a unital homomorphism $\mathbb{Q} \to \mathbb{F}_5$.
Hint 1 — a starting point. Consider the inverse of five in $\mathbb{Q}$.
Hint 2 — the key move. A homomorphism would send $5 \cdot \left(\frac{1}{5}\right) = 1$ to zero.

MM5020-U7.09 · C · 30 min
Learning objective: U7.2 Root adjunction
Construct a field containing a root of x^3+x+1 over $\mathbb{F}_2$ and list a basis over $\mathbb{F}_2$.
Hint 1 — a starting point. First check irreducibility using the two possible roots.
Hint 2 — the key move. Use the quotient by the polynomial.

MM5020-U7.10 · A · 50 min
Learning objective: U7.1 Compositum degree
Prove $\left[\mathbb{Q}\left(\sqrt{2},\sqrt{5}\right):\mathbb{Q}\right]=4$ and find a basis.
Hint 1 — a starting point. Show $\sqrt{5}$ is not in $\mathbb{Q}\left(\sqrt{2}\right)$.
Hint 2 — the key move. Square a proposed expression $a+b\sqrt{2}$.

MM5020-U7.11 · A · 50 min
Learning objective: U7.2 Transitivity of algebraicity
Let L/K be algebraic and let α be algebraic over L. Prove α is algebraic over K, even when L/K is infinite.
Hint 1 — a starting point. Only finitely many coefficients occur in an annihilating polynomial.
Hint 2 — the key move. Adjoin those coefficients to K first.

MM5020-U7.12 · A · 50 min
Learning objective: U7.2 A primitive generator
Show $\mathbb{Q}\left(\sqrt{2}+\sqrt{3}\right)=\mathbb{Q}\left(\sqrt{2},\sqrt{3}\right)$ and find the minimal polynomial of the sum.
Hint 1 — a starting point. The reciprocal of the sum recovers the difference.
Hint 2 — the key move. Since $\left(\sqrt{3}-\sqrt{2}\right)\left(\sqrt{3}+\sqrt{2}\right)=1$, both roots can be recovered.

MM5020-U7.13 · A · 60 min · Optional challenge

Learning objective: U7 Transfer: A field containing three roots

Let a be the positive real cube root of two and let ω be a nonreal root of $x^2 + x + 1$. Determine the smallest subfield of $\mathbb{C}$ containing every root of $x^3 - 2$, and its degree over $\mathbb{Q}$.

Hint 1 — a starting point. List the three roots and compare their ratios.

Hint 2 — the key move. The field $\mathbb{Q}(a)$ is real, while ω satisfies a quadratic.

8 Unit 8 · Finite fields and applications

MM5020-U8.01 · E · 15 min
Learning objective: U8.1 Field orders
Which of the orders $8, 9, 10, 16, 18, 25$ can occur for a finite field?
Hint 1 — a starting point. Finite-field orders are prime powers.
Hint 2 — the key move. Apply the existence theorem as well as the necessary condition.

MM5020-U8.02 · E · 15 min
Learning objective: U8.1 Arithmetic in four elements
In $\mathbb{F}_4 = \mathbb{F}_2[\alpha]$ with $\alpha^2 + \alpha + 1 = 0$, compute $\alpha(\alpha + 1)$ and the inverse of $\alpha + 1$.
Hint 1 — a starting point. Use $\alpha^2 = \alpha + 1$.
Hint 2 — the key move. Addition has characteristic two.

MM5020-U8.03 · E · 15 min
Learning objective: U8.2 Generators
Find the order of two in $\mathbb{F}_7^{\times}$.
Hint 1 — a starting point. Compute successive powers.
Hint 2 — the key move. The third power is one.

MM5020-U8.04 · E · 15 min
Learning objective: U8.1 Characteristic and dimension
What are the characteristic and dimension over the prime field of a field of order eighty-one?
Hint 1 — a starting point. Write the order as a prime power.
Hint 2 — the key move. Eighty-one equals three to the fourth power.

MM5020-U8.05 · C · 30 min
Learning objective: U8.2 Frobenius
Prove the map $a \mapsto a^p$ is an automorphism of a finite field of characteristic p.
Hint 1 — a starting point. Check addition, multiplication, and injectivity.
Hint 2 — the key move. Use the binomial theorem and the fact a field has no nonzero nilpotents.

MM5020-U8.06 · C · 30 min
Learning objective: U8.2 Interpolation
Find the degree-less-than-three polynomial over $\mathbb{F}_5$ with $f(0) = 1$, $f(1) = 0$, $f(2) = 2$.
Hint 1 — a starting point. Write $f = ax^2 + bx + c$.
Hint 2 — the key move. The first value gives $c = 1$; solve the remaining two equations modulo five.

MM5020-U8.07 · C · 30 min
Learning objective: U8.1 Polynomial roots
Show every element of $\mathbb{F}_q$ is a root of $x^q - x$, and this polynomial has no repeated root.
Hint 1 — a starting point. Apply Lagrange to the nonzero elements.
Hint 2 — the key move. Its formal derivative is negative one.

MM5020-U8.08 · C · 30 min
Learning objective: U8.2 Evaluation codes
Evaluate all degree-less-than-two polynomials at the five elements of $\mathbb{F}_5$. How many codewords result, and what is their minimum distance?
Hint 1 — a starting point. There are two free field coefficients.
Hint 2 — the key move. A nonzero difference is a nonzero polynomial of degree at most one.

MM5020-U8.09 · C · 30 min
Learning objective: U8.2 Primitive elements
Find all generators of $\mathbb{F}_7^\times$, given that three is one generator.
Hint 1 — a starting point. In a cyclic group of order six, generator powers have exponents coprime to six.
Hint 2 — the key move. Use exponents one and five.

MM5020-U8.10 · A · 50 min
Learning objective: U8.1 Subfields
Show a field of sixteen elements has a unique subfield of four elements.
Hint 1 — a starting point. Use the roots of $x^4 - x$ inside the larger field.
Hint 2 — the key move. Its nonzero roots satisfy $x^3 = 1$ in a cyclic group of order fifteen.

MM5020-U8.11 · A · 50 min
Learning objective: U8.2 Polynomial functions
Prove every function $\mathbb{F}_q \to \mathbb{F}_q$ is represented by a unique polynomial of degree less than q. Is the representing polynomial unique without the degree bound?
Hint 1 — a starting point. Interpolate at every element of the field.
Hint 2 — the key move. The polynomial $x^q - x$ represents the zero function.

MM5020-U8.12 · A · 50 min
Learning objective: U8.1 Irreducible quadratics
Count the monic irreducible quadratics over $\mathbb{F}_q$.
Hint 1 — a starting point. Count all monic quadratics and subtract the reducible ones.
Hint 2 — the key move. Reducible monic quadratics correspond to unordered pairs of roots, allowing repetition.

MM5020-U8.13 · A · 60 min · Optional challenge

Learning objective: U8 Transfer: An equation over a larger finite field

For each $b \in \mathbb{F}_q$, determine the number of $x \in \mathbb{F}_{q^2}$ satisfying $x^q + x = b$. Include the case of characteristic two.

Hint 1 — a starting point. Show that the left side always lies in the subfield $\mathbb{F}_q$.

Hint 2 — the key move. Use linearity over $\mathbb{F}_q$ and the root bound to show the map is not zero.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM5020-M1.01 · E · 15 min
Learning objective: U1–U2 Cyclic groups
List the orders of every element of $\mathbb{Z}/8\mathbb{Z}$.
Hint 1 — a starting point. Use the gcd formula.
Hint 2 — the key move. For residue k, the order is $\frac{8}{\gcd(8,k)}$.

MM5020-M1.02 · E · 15 min
Learning objective: U5 Ring quotients
Determine whether $\mathbb{Z}/15\mathbb{Z}$ is a field and list its units.
Hint 1 — a starting point. Check coprimality to fifteen.
Hint 2 — the key move. The product of three and five is zero.

MM5020-M1.03 · C · 30 min
Learning objective: U3–U4 Group structure
Let $|G| = 49$. Prove G is abelian, and show that if some element has order forty-nine then G is cyclic.
Hint 1 — a starting point. Use the center of a prime-power group.
Hint 2 — the key move. The center has order seven or forty-nine.

MM5020-M1.04 · C · 30 min
Learning objective: U6–U7 Algebraic extensions
Let $\alpha^3 = 3$ over $\mathbb{Q}$. Prove the extension degree is three and find $(\alpha - 1)^{-1}$.
Hint 1 — a starting point. Use Eisenstein and a geometric-product identity.
Hint 2 — the key move. The product with $\alpha^2 + \alpha + 1$ is two.

MM5020-M1.05 · A · 40 min
Learning objective: U3 Universal constructions
For normal subgroups N, M of G, show $g \mapsto (gN, gM)$ induces an injective homomorphism $G/(N \cap M) \to G/N \times G/M$. When is it surjective?
Hint 1 — a starting point. Compute the kernel of the paired map.
Hint 2 — the key move. Surjectivity is equivalent to $G = NM$.

MM5020-M1.06 · A · 40 min
Learning objective: U8 Field structure
Prove that the fixed field of Frobenius $x \mapsto x^p$ in $\mathbb{F}_{p^n}$ is exactly $\mathbb{F}_p$.
Hint 1 — a starting point. Fixed elements are roots of $x^p - x$.
Hint 2 — the key move. The prime field already supplies p roots.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM5020-M2.01 · E · 15 min
Learning objective: U2 Permutations
Find the order and sign of $(123)(45)(67)$.
Hint 1 — a starting point. Use disjoint cycles.
Hint 2 — the key move. Take the least common multiple of three, two, and two.

MM5020-M2.02 · E · 15 min
Learning objective: U7 Minimal polynomials
Find the minimal polynomial of $2+\sqrt{5}$ over $\mathbb{Q}$.
Hint 1 — a starting point. Isolate the square root.
Hint 2 — the key move. Use $(x-2)^2-5$.

MM5020-M2.03 · C · 30 min
Learning objective: U3 Quotient isomorphisms
Prove $S_4/A_4 \cong C_2$.
Hint 1 — a starting point. Use sign.
Hint 2 — the key move. Its kernel is A_4 and a transposition maps to negative one.

MM5020-M2.04 · C · 30 min
Learning objective: U5–U6 Field construction
Prove $\mathbb{F}_3[x]/(x^2+1)$ is a field and find the inverse of $[x+1]$.
Hint 1 — a starting point. Check roots, then multiply by $x-1$.
Hint 2 — the key move. Modulo x^2+1, the product is negative two, which equals one in $\mathbb{F}_3$.

MM5020-M2.05 · A · 40 min
Learning objective: U4 Actions
A group G of order fifteen acts on a set of seven elements. Prove there is a fixed point.
Hint 1 — a starting point. Orbit sizes divide fifteen.
Hint 2 — the key move. Only sizes one, three, and five fit inside seven.

MM5020-M2.06 · A · 40 min
Learning objective: U7–U8 Degrees and subfields
Prove that $\mathbb{F}_{p^3}$ has no subfield strictly between $\mathbb{F}_p$ and itself.
Hint 1 — a starting point. Translate subfield cardinalities into vector-space degrees.
Hint 2 — the key move. The tower law makes the intermediate degree divide three.